

ASSIGNMENT – 16

ADVANCED PROGRAMMING

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SESSION-SPRING

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```
producer_consumer_windows.c > Producer(LPVOID)
1  #include <stdio.h>
2  #include <windows.h>
3
4  #define BUFFER_SIZE 5
5  #define ITEMS 10
6
7  int buffer[BUFFER_SIZE];
8  int count = 0;
9
10 // Synchronization objects
11 HANDLE mutex;
12 HANDLE emptySlots;
13 HANDLE filledSlots;
14
15 // Producer thread
16 DWORD WINAPI Producer(LPVOID arg) {
17     for (int i = 1; i <= ITEMS; i++) {
18
19         // Wait for empty slot
20         WaitForSingleObject(emptySlots, INFINITE);
21
22         // Lock buffer
23         WaitForSingleObject(mutex, INFINITE);
24
25         buffer[count] = i;
26         count++;
27
28         printf("Producer produced: %d | Buffer count: %d\n", i, count);
29
30         // Unlock buffer
31         ReleaseMutex(mutex);
32
33         // Signal filled slot
34         ReleaseSemaphore(filledSlots, 1, NULL);
35
36         Sleep(500);
37     }
```

```
producer_consumer_windows.c > Producer(LPVOID)
16  DWORD WINAPI Producer(LPVOID arg) {
17      for (int i = 1; i <= ITEMS; i++) {
34          ReleaseSemaphore(filledSlots, 1, NULL);
35
36          Sleep(500);
37      }
38      return 0;
39  }
40
41  // Consumer thread
42  DWORD WINAPI Consumer(LPVOID arg) {
43      int item;
44
45      for (int i = 1; i <= ITEMS; i++) {
46
47          // Wait for filled slot
48          WaitForSingleObject(filledSlots, INFINITE);
49
50          // Lock buffer
51          WaitForSingleObject(mutex, INFINITE);
52
53          item = buffer[count - 1];
54          count--;
55
56          printf("Consumer consumed: %d | Buffer count: %d\n", item, count);
57
58          // Unlock buffer
59          ReleaseMutex(mutex);
60
61          // Signal empty slot
62          ReleaseSemaphore(emptySlots, 1, NULL);
63
64          Sleep(700);
65      }
66      return 0;
67  }
68
```

```

1 producer_consumer_windows.c > Producer(LPVOID)
42  DWORD WINAPI Consumer(LPVOID arg) {
43      for (int i = 1; i <= ITEMS; i++) {
44      }
45      return 0;
46  }
47
48  int main() {
49      HANDLE producerThread, consumerThread;
50
51      // Create synchronization objects
52      mutex = CreateMutex(NULL, FALSE, NULL);
53      emptySlots = CreateSemaphore(NULL, BUFFER_SIZE, BUFFER_SIZE, NULL);
54      filledSlots = CreateSemaphore(NULL, 0, BUFFER_SIZE, NULL);
55
56      // Create threads
57      producerThread = CreateThread(NULL, 0, Producer, NULL, 0, NULL);
58      consumerThread = CreateThread(NULL, 0, Consumer, NULL, 0, NULL);
59
60      // Wait for threads to finish
61      WaitForSingleObject(producerThread, INFINITE);
62      WaitForSingleObject(consumerThread, INFINITE);
63
64      // Cleanup
65      CloseHandle(producerThread);
66      CloseHandle(consumerThread);
67      CloseHandle(mutex);
68      CloseHandle(emptySlots);
69      CloseHandle(filledSlots);
70
71      printf("\nExecution completed successfully.\n");
72
73      return 0;
74  }
75
76

```

OUTPUT-

```

c:\Users\VITU\OneDrive\Desktop>cd "c:\Users\VITU\OneDrive\Desktop\c\" && gcc producer_consumer_windows.c -o producer_consumer_windows && "c:\Users\VITU\OneDrive\Desktop\c\"producer_consumer_windows
Producer produced: 1 | Buffer count: 1
Consumer consumed: 1 | Buffer count: 0
Producer produced: 2 | Buffer count: 1
Consumer consumed: 2 | Buffer count: 0
Producer produced: 3 | Buffer count: 1
Consumer consumed: 3 | Buffer count: 0
Producer produced: 4 | Buffer count: 1
Producer produced: 5 | Buffer count: 2
Consumer consumed: 5 | Buffer count: 1
Producer produced: 6 | Buffer count: 2
Consumer consumed: 6 | Buffer count: 1
Producer produced: 7 | Buffer count: 2
Consumer consumed: 7 | Buffer count: 1
Producer produced: 8 | Buffer count: 2
Producer produced: 9 | Buffer count: 3
Consumer consumed: 9 | Buffer count: 2
Producer produced: 10 | Buffer count: 3
Consumer consumed: 10 | Buffer count: 2
Consumer consumed: 8 | Buffer count: 1
Consumer consumed: 4 | Buffer count: 0

Execution completed successfully.

c:\Users\VITU\OneDrive\Desktop>

```

